KUMBHATHON - NASHIK

Atreya

Outbreak early warning for mass gatherings

Every clinic treats its own patients well.
Nobody compares notes. We do that, every five minutes.

THE ONE-PARAGRAPH VERSION

Millions of people gather. Fourteen of them drink from a cracked pipe and go to four different clinics at four different hours. Each doctor sees three or four cases of diarrhoea, which is a completely normal Tuesday, so nobody raises an alarm - and there was no alarm to raise, because the number fourteen did not exist anywhere. It was three here and four there, in four separate paper registers that get added up in November. **Everyone got treated. Nobody turned off the tap.**

Atreya is the ten minutes after the data is entered. Doctors log each visit in ten seconds on a phone that works with no network. Every five minutes we compare each area against its own recent history, at the same time of day, and flag what is genuinely unusual - with a z-score, not machine learning, so a health officer can ask why it fired and get an answer they can argue with. The alert becomes a job with a named crew who must close it. In testing across a thousand synthetic festivals we catch a typical waterborne outbreak at **8 cases instead of 33**, a median of **seven hours** earlier.

Live: health.akhileshnagargoje.in
Engine: health-engine.akhileshnagargoje.in

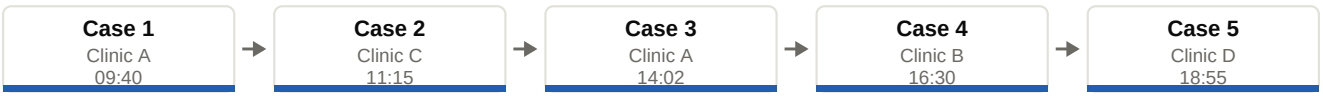
Source: github.com/AkhileshNagargoje/siddhi
Built for the 2027 Nashik Kumbh Mela

01 - THE PROBLEM

Nobody is doing anything wrong

This is the part that is easy to get wrong when you explain it. The failure is not bad medicine, missing clinics, or lazy staff. Every single patient in the story above received correct treatment. The failure is that the **information never adds up anywhere**.

Why an outbreak is invisible for days



Five cases, four clinics, nine hours apart. No single doctor sees more than two. Each one is a normal day at that tent. The cluster exists only when somebody adds the four registers together - and that happens in November.

By day four it is no longer a detection problem, it is a logistics problem.

Four hundred cases. A five-hour queue. And pilgrims boarding trains home to forty districts, carrying it with them. The cost of the delay is not the first fourteen people. It is everyone infected during the days nobody knew.

What already exists, and what it stops short of

In 2015 a Harvard-led project ran a digital surveillance experiment at this exact festival - the Jana Swasthya project. It proved the hard part: health workers **will** enter data on tablets, at volume, in a field tent. Over 2,000 visits were recorded on day one across more than 30 clinics.

And then it drew charts with it. Its own published data showed antibiotics prescribed in up to **31% of encounters** - recorded faithfully, and never acted on. The data collection was solved a decade ago. The loop from data to a person walking somewhere was not.

Our sentence, if you only remember one.

They proved the data can be collected. We built what happens next: see it, detect it, dispatch it, and carry it home.

02 - WHAT WE BUILT

Four screens, one loop

Atreya is not one app. It is four narrow screens, each built for one person with one job, joined by a detection engine in the middle. Nobody is asked to learn a system.

WHO	SCREEN	WHAT THEY DO	DESIGN CONSTRAINT
Doctor / health worker	/entry	Log a visit: age band, sex, symptoms, diagnosis, medicine, severity.	Must never block the doctor. Measured at 2.2 seconds. Works with no network.
Medicine stall	/stall	Tally units sold - ORS, zinc, antidiarrheal.	People self-medicate before they consult. This signal leads the clinic one.
Health official	/dashboard	Watch 12 sectors live, open an alert, read its arithmetic, dispatch a crew.	Every alert shows observed, expected and how many noise-widths above. Arguable by design.
Field crew	/crew	Acknowledge the job, close it with what they actually found.	Closing the loop is what makes time-to-response measurable at all.
Pilgrim	/public	Scan a printed QR. See their area, the illness going around, what to do.	No app, no login, three languages - and never a case count. See section 07.

The loop, end to end

1 ENTER	2 DETECT	3 ALERT	4 DISPATCH	5 CLOSE
doctor taps 10 seconds	every 5 min z-score	observed vs expected	named crew must ack	outcome + 4 timestamps

Four timestamps - detected, dispatched, acknowledged, closed - are written by the system, not self-reported. That is why "time to response" is a number on the screen rather than a claim in a report. Nobody can round it.

The second signal

Clinic encounters are one stream. Medicine-stall sales are another, and they are genuinely independent: a different reporter, a different act (a purchase, not a consultation), and different failure modes. Rehydration sales rising in a sector is corroboration - and it catches the people who never reach a clinic at all, which is the single biggest blind spot in clinic-only surveillance.

> **Say this before anyone asks.**
>
> In the demo both streams are simulated and one command writes both. What you are seeing is the architecture, not the evidence for it. If the two streams disagree in the field, that is the more interesting case: clinic cases up with no rehydration spike suggests it may not be waterborne, which changes where you send the crew first.

03 - THE DETECTION MATHS

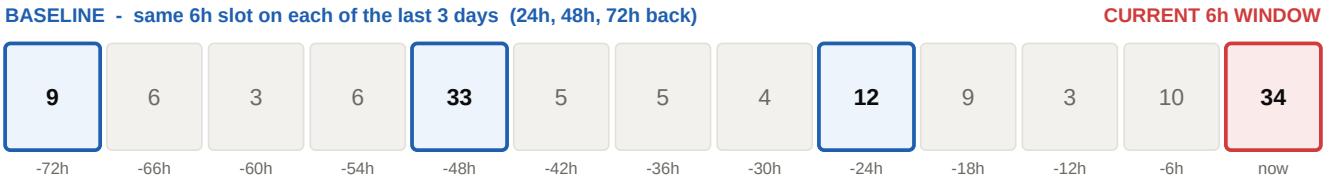
Small enough to check by hand

The detector is deliberately tiny. A district health officer will act on "fourteen against four expected" and will ignore a black box, so explainability here is an adoption feature, not a limitation.

Step 1 - cut the festival into cells

12 sectors x 6 syndromes (gi, respiratory, heat, fever, injury, other) = **72 cells**. Each is scored independently, every five minutes.

Step 2 - compare each cell against its own past, at the same time of day



One cell = one sector, one syndrome, six hours. 12 sectors x 6 syndromes = 72 cells, rescored every 5 minutes.

Why the same time of day, and not simply "the last three days"? Clinics are busy in the morning and quiet at 3am. Averaged against the night windows, **every single afternoon looks like an outbreak**. Comparing an afternoon only with previous afternoons removes the daily rhythm without modelling it.

Step 3 - score it

$$z = \frac{\text{observed} - \text{baseline mean}}{\max(\text{standard deviation}, \sqrt{\text{mean}}, 1)}$$

Fires when z >= 3 AND observed >= 8. Tier "act" (crew dispatched) at z >= 4.

Why each piece is there

TERM	WHY IT EXISTS

observed - mean	How far above its own normal this cell is. Not above the festival average - a busy sector is allowed to be busy.
standard deviation	A cell that swings wildly day to day needs a bigger jump to be surprising. A steady cell does not.
sqrt(mean) - the Poisson floor	Three quiet days in a row give a standard deviation of zero, which makes z infinite and every cell scream. Count data has irreducible noise of about sqrt(mean), so the divisor is never allowed below it. This one term kills most small-sample false alarms.
the final 1	Covers a mean of zero, where even sqrt(mean) is zero. Prevents division by zero.
observed >= 8	Without it, one case against a baseline of zero scores infinitely. Eight is the smallest count worth a human walking somewhere.
z >= 3	A triage threshold, not a p-value. Lower catches marginally more and buries the dashboard; higher saves few false alarms and costs hours of lag. Three is where the curve bends - and the harness sweeps it, so the trade is shown rather than asserted.

> **If a judge says "three sigma, so a one-in-740 event?" - do not say yes.**
>
> It is a triage threshold, not a tail probability. It means this cell is at least three Poisson noise-widths above what the same six-hour slot looked like on the last three days. It is tuned to put roughly the right number of things in front of a human, which is exactly why we measured the false-alarm rate instead of quoting a p-value.

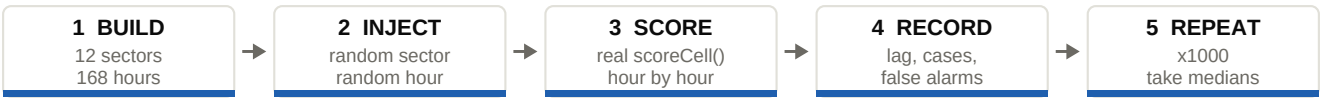
04 - THE EVALUATION HARNESS

We measured it instead of asserting it

This is the strongest thing we have, and almost no hackathon team brings one. A flight simulator for the detector: it builds a thousand synthetic festivals, injects an outbreak at a random sector and a random hour in half of them, runs the **real deployed detection function** over each hour, and records what happened.

```
cd worker && node eval/harness.mjs --trials 1000 --seed 42
```

What one trial does

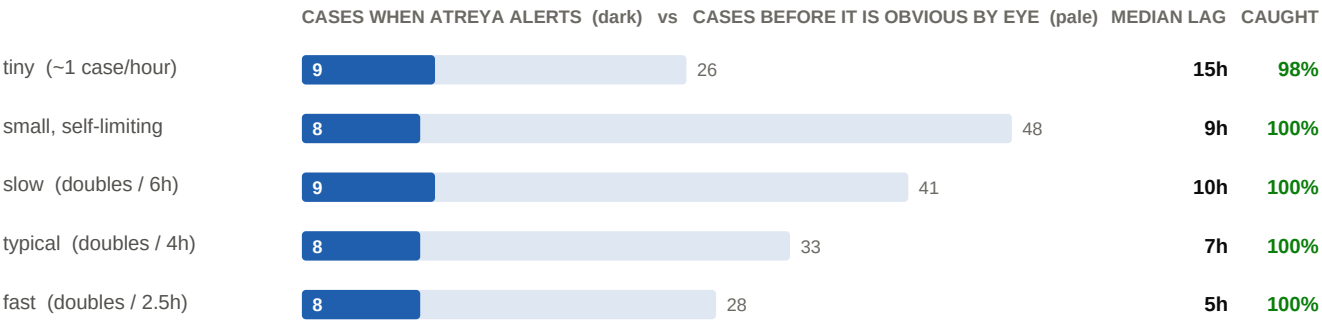


168 hours = 72 hours of warm-up so the baseline has history, plus 96 hours actually evaluated. Hourly counts are Poisson draws around a rate set by the number of clinics in that sector, the syndrome mix, and the time of day. Outbreaks grow exponentially from a point source: rate = a0 x 2^(hours/doubling).

Four design choices that make the numbers trustworthy

CHOICE	WHY IT MATTERS
It imports the deployed algorithm	The harness begins with `import { scoreCell } from '../src/index.js'` - the same function the live Worker calls. Not a reimplementation that could quietly drift. Change the detector and these numbers change.
It is deterministic	Seeded generator. `--seed 42` produces identical numbers on any machine. That is the difference between a measurement and an anecdote.
It mirrors the engine's dedup	After a cell fires it goes quiet until its window turns over, exactly as the live system does. Without this, one outbreak would count as six alarms and our numbers would flatter us.
False alarms come from clean festivals	Nothing is injected at all. This half cannot be gamed by construction - we cannot accidentally tune ourselves to detect an outbreak that is not there.

Results



1,000 synthetic festivals per row, seed 42, 96h evaluated. Medians, not means - outbreak latency has a long tail.

False alarms, reported separately because they cost different things

TIER	WHAT HAPPENS	MEAN / DAY	p90
act	A crew is dispatched. This is the expensive one.	**1.5**	2.3
watch	Appears on the dashboard. Nobody is sent anywhere.	**5.4**	6.8

The caveat we volunteer, because it is the honest number.

Those figures come from traffic generated as a pure Poisson draw. Our own live simulator is overdispersed - it multiplies each hourly mean by a random factor between 0.6 and 1.4 - and against that burstier traffic the detector fires considerably more often. Neither model is wrong; they describe different worlds. The honest claim is "6.8 a day under a Poisson traffic model, more under bursty traffic - the real rate depends on how bursty real festival traffic is." We would rather say that than quote the friendlier number and be caught.

05 - THE SIMULATOR

Fake data, real pipe

There is no live Kumbh to plug into, so we generate the traffic. The single rule that makes this defensible is in our project spec: **the simulator must write through the same public API a real tablet writes to.** It has no special access and no direct database connection. The detector cannot tell the difference, because there is no difference to tell.

COMMAND	WHAT IT DOES
node simulate.js backfill	~4,900 encounters across 84 hours with the real daily rhythm. Run this after any baseline reset - the detector needs history to compare against.
node simulate.js live	A trickle of new patients, so the board is never a dead screen while you talk.
node simulate.js outbreak 5	A point-source waterborne cluster in one sector, growing wave by wave.
node simulate.js pharmacy	The independent medicine-sales stream.

How the traffic is shaped

Six patients per clinic per peak hour, multiplied by a time-of-day factor (busy mornings, near-dead nights), multiplied by a random 0.6-1.4. Age bands follow a realistic festival mix (5% under-five, 25% over-sixty), and the case mix is weighted across the six syndromes.

"Your data is fake." - Say it first, before they do.

Yes. It is synthetic, generated through the same API a real tablet writes to, so the detector cannot distinguish it. That is precisely what let us measure detection performance across a thousand festivals instead of asserting it once. The alternative - a small pile of real data - would let us claim less, not more.

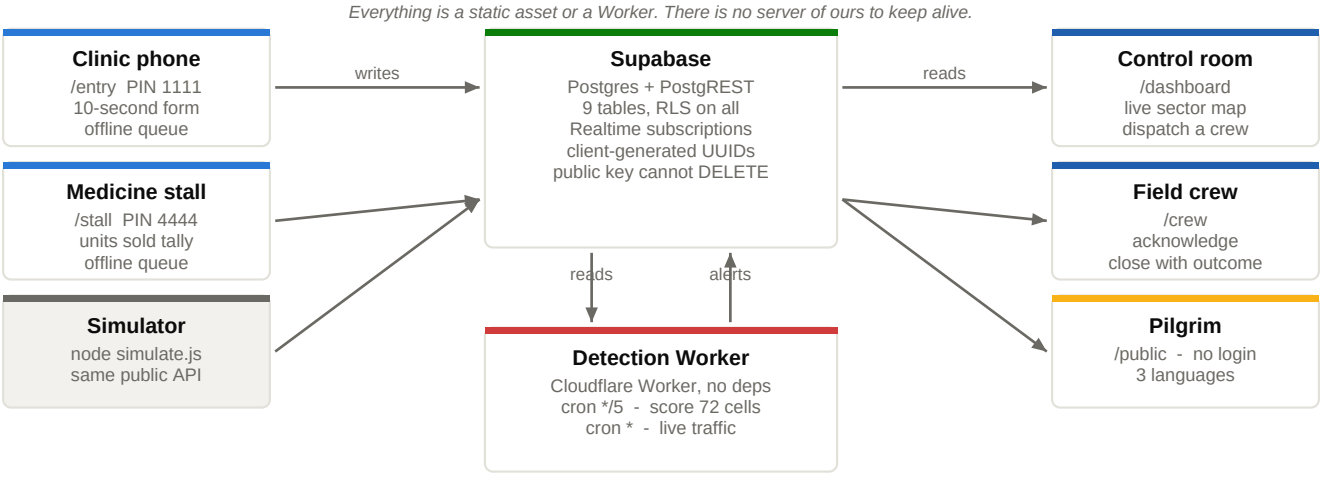
One weakness we found by breaking it ourselves

Repeated demo injections into one sector wrote a fake spike into that sector's own history. Because the baseline is built from the same slot on previous days, the detector correctly concluded that a spike is normal there - and sector 5 reached a state where it needed roughly **295 cases in six hours** before it would say anything. It had gone deaf.

This is the documented "one noisy historical day desensitises the cell" weakness, and it is written into our own unit tests as accepted behaviour. We mention it because a limitation you found yourself and can reproduce on demand is stronger evidence of rigour than a limitation you were told about. Recovery is `db/reset-baseline.sql` followed by a fresh backfill.

06 - ARCHITECTURE AND STACK

Nothing of ours to keep alive



The stack, and why each piece

LAYER	CHOICE	WHY
Frontend	React 19 + Vite 8, code-split lazy routes	No UI framework, no CSS framework, no state library. The pilgrim page is a few KB because it may open on a cheap phone on a bad network.
Data	Supabase (Postgres, PostgREST, Realtime)	The dashboard subscribes to changes rather than polling. Nine tables, Row Level Security on all of them.
Engine	A second Cloudflare Worker, zero dependencies	Two crons: detection every 5 minutes, live traffic every minute. Plain JavaScript - the whole scoring function fits on one screen.
Hosting	Cloudflare Workers static assets	No server of ours to keep alive during a festival, and nothing to scale by hand on a bathing day.
Auth	PIN codes. Deliberately not real auth.	Demo theatre. A real deployment would use the state health department's own identity, not ours.

Offline-first, which matters more than anything else we built

If the entry form is slow or fails on a bad network, doctors abandon it by day three and every other screen goes dark. So the phone never waits for the network:

- The row is written to a local queue and the form clears immediately.
- Every record carries a **client-generated UUID**, so the phone owns the identity of the row before the server has ever seen it.
- Sync upserts with `ignoreDuplicates`, and a duplicate-key error (23505) is treated as **success** - because it means the row is already safely there.
- Only the IDs actually confirmed sent are removed from the queue. A half-failed sync loses nothing.

The consequence: a doctor in a tent with no signal keeps working, and the control room gets everything the moment a bar of signal appears. Nobody has to think about it.

07 - THE PILGRIM SCREEN

Where the safety thinking is

This screen is reached by scanning a printed poster. No app, no login, nothing recorded. It shows which areas are affected and which illness is going around - and it **never shows a case count**.

> **This is a design decision, not squeamishness.**
>
> A number invites comparison between areas. Comparison moves crowds. And crowd movement is what kills at these events - the 2025 Maha Kumbh crush is the reference. So every status ends in an instruction, because an instruction calms and a bare warning does not, and the board says plainly that a marked area is not a reason to move, because every area has its own free clinic.

Three levels, not seven

The health official's sector map is a seven-step heat ramp keyed to case counts, with a "fewer to more cases" legend. The pilgrim map keeps the shape and swaps the quantity: three levels keyed to **what to do**, because there are only three things to do. A count ramp with the digits removed is still a count ramp - a reader infers the number from the colour.

LEVEL	WHEN	WHAT THE PILGRIM SEES
Take extra care	an act-tier alert - a crew was dispatched	Red tile, 8px bar, the illness named, targeted advice.
Take care	watch-tier only - being observed	Amber tile, 4px bar, the illness named.
All clear	nothing flagged	Plain tile, no bar.

The level is the same tier the health official sees, so the two screens can never contradict each other. The grid stays in fixed order rather than sorted worst-first, because a leaderboard is an instruction to move - and because a returning reader finds their own area in the same place every time.

Advice aimed at whoever is actually falling ill

When one vulnerable group is over-represented among a sector's cases, the page shows advice written for that group - keep feeding the child, ORS is free, and the one symptom that means go now. The targeting is **share-based, not count-based**: 18-39 is the largest band at any festival, so "most cases" would point there almost every time and tell a parent nothing. A group is named only when its share of this sector's cases clearly exceeds its share everywhere else, and only when enough cases back it.

Everything is in English, Hindi and Marathi.

> **State this if asked - it is a strength, not an admission.**
>
> The Hindi and Marathi have not been verified by a native speaker, and the symptom wording follows standard public-health guidance but has not been reviewed by a clinician. Both are required before this is shown to a real pilgrim, and both are noted in the source file itself. We would rather ship a known gap than an unmarked one.

The referral card

A pilgrim found to have high blood pressure at a festival clinic should not lose that finding on the train home. The printed QR carries **the record itself, not a link** - so a village clinic in another state reads it with no network, no app and no ID system. Paper for the patient, digital for the system.

08 - WHAT WE DID NOT BUILD

The questions we would ask us

Every one of these is a real limitation. Conceding them immediately is worth more than defending them badly - and each one has an answer.

"What about a bathing day, when the crowd triples?"

Correct, and it is the first thing we would fix. We detect on raw counts, so a genuine crowd surge would false-positive across every sector at once. The fix is rates per thousand people present, using the crowd estimates the administration already produces. It is designed and not built - we chose to make what exists solid rather than half-build a second thing.

"Why no AI?"

Deliberately. A district health officer will act on "fourteen against four expected, four noise-widths above" and will ignore a black box - explainability is an adoption feature here. And practically: 72 series with no labelled outbreak history is nothing for a model to learn from. There is no signal arithmetic does not already give you.

"Why not scan Aadhaar and pull the patient's history?"

Three reasons, and the first settles it. **Aadhaar does not hold medical history** - authentication returns demographics, so scanning it would give us a name we deliberately do not want and none of the clinical history you are imagining. The system that does hold health records is ABHA, under the Ayushman Bharat Digital Mission.

Second, we could not do it lawfully anyway: since the 2025 rules a private entity must route each use case through a ministry to UIDAI. That is a governmental process, not an integration.

Third, and this is the real answer - **identity adds nothing to what we do**. We count cases per sector against that sector's own history. A z-score does not care who anyone is. Adding identity would cost us the ten-second entry, and at a Kumbh it would turn away exactly the elderly and rural pilgrims whose fingerprints fail authentication - the people most at risk. The one thing identity would genuinely buy, a finding following the patient home, we solved with the printed QR card instead.

"Isn't this just a dashboard?"

A dashboard is where this stops for everyone else. Ours turns every alert into an assigned job that a named crew has to close with an outcome - which is why we can measure time-to-response at all. That is the whole difference.

Honest status of the rest

AREA	STATUS
Crowd normalisation	Designed, not built. The bathing-day answer above.
Real-world validation	Not done. Every number here is from synthetic data. The highest-value next step is running the detector against the public Jana Swasthya dataset (Harvard Dataverse, DOI 10.7910/DVN/WCEWNR).
Clinical review	Not done. Advice wording and the antibiotic table both need a clinician.
Translations	Not verified by native speakers.
Authentication	PINs only. A real deployment uses the department's own identity.
Baseline poisoning	Known and documented. One noisy historical day desensitises a cell for 72 hours.

09 - THINGS THAT GET LEFT OUT

Say these - they are the differentiators

Under time pressure the demo swallows the story and these get dropped. Each one is a reason to prefer this project over a prettier dashboard.

#	THE POINT	THE SENTENCE
1	**We measured it**	"A thousand synthetic festivals, running the deployed function, seeded so anyone can reproduce it. We can show you what a different threshold would have cost." Almost nobody brings an evaluation.
2	**Two independent signals**	"Clinic visits and medicine sales. Different reporter, different act, different failure modes. Two signals agreeing is evidence; one signal shouting twice is not."
3	**It closes the loop**	"Every alert becomes a job a named crew must close with an outcome. That is why time-to-response is a measured number and not a claim."
4	**Offline is the product**	"If the form is slow on a bad network, doctors abandon it by day three and every other screen goes dark. The phone never waits for the network."
5	**Explainable on purpose**	"Every alert shows its own arithmetic. An officer can argue with it - and they ignore anything they cannot argue with."
6	**Crowd safety in the UI**	"The pilgrim screen never shows a case number, because numbers invite comparison and comparison moves crowds. That is a design constraint we imposed on ourselves."
7	**Config-driven**	"Sectors, clinics, syndromes and thresholds are config. Point it at a different gathering by editing one file - it is not hardcoded to Nashik."
8	**We broke it ourselves**	"We poisoned our own baseline with repeated demo injections and made a sector go deaf. We can reproduce it, and the recovery is in the repo."
9	**The record travels**	"The QR carries the record, not a link. A village clinic reads it with no network and no app."
10	**We did not over-build**	"Tier 1 and Tier 2 are complete and deployed. Crowd normalisation is designed and deliberately not built. We would rather ship four solid screens than eight half-finished ones."

10 - NUMBERS AND SOURCES

Every figure, and where it comes from

If a number is not on this page, do not say it on stage.

FIGURE	MEANING	SOURCE
8 vs 33 cases	Cluster size when Atreya alerts, against size before it is obvious by eye.	worker/eval/RESULTS.txt - reproducible with node eval/harness.mjs --trials 1000
7 hours	Median detection lag, typical waterborne outbreak.	Same harness run.
100% caught	Typical, fast, slow and self-limiting outbreaks. Tiny outbreaks: 98%.	Same harness run.
1.5 / day	Act-tier false alarms - a crew is dispatched.	Clean festivals, nothing injected. See the Poisson caveat in section 04.
5.4 / day	Watch-tier false alarms - dashboard only, nobody sent.	Same.
2.2 seconds	Measured time to log one patient on the entry form.	Timed on the deployed app.
31%	Antibiotic prescribing rate recorded at the 2015 Kumbh, never acted on.	Shaikh et al., JMIR Public Health and Surveillance, 2023
2,000+ visits, 30+ clinics	Jana Swasthya day one - proof that health workers will enter data.	Harvard Mittal South Asia Institute

> **One number we deliberately do not use.**
>
> An earlier draft claimed "600 cases reduced to 35". It is a logistic curve fitted to nothing, and it would sit in the same seven minutes as the measured 33-to-8 from the harness. Two counterfactuals an order of magnitude apart invites the one question we could not answer well. The harness owns every number in this deck.

Reproduce any of it

```
git clone https://github.com/AkhileshNagargoje/siddhi
cd worker && npm install && node eval/harness.mjs
cd worker && npm test
```
(unit tests for the scoring function)

Atreya - built for Kumbhathon, Nashik. Live at health.akhileshnagargoje.in. Full technical documentation, the detection spec, the evaluation harness and a complete handover document are in the repository.